Design and Analysis of a Low-Power CCFF for VLSI Integrated Circuits

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INTRODUCTION

A Conditional/Clocked Controlled Flip-Flop (CCFF) was designed in this project, which is an essential sequential building block in VLSI integrated circuits. The purpose of this design is to reduce power consumption by minimizing unnecessary internal switching through conditional clock operation.

The circuit is implemented at the transistor level using CMOS technology, focusing on power efficiency while maintaining correct functionality and reliable performance.

Such flip-flop designs are widely used in integrated circuit systems, including processors, control units, and low-power VLSI applications.

OUR PROJECT

We designed a low-power Conditional-Capture Flip-Flop (CCFF) in 32nm CMOS technology, focusing on eliminating redundant internal transitions to achieve significant power savings. Our project encompasses complete schematic design, physical layout, and comprehensive pre-layout and post-layout simulations. The CCFF achieves 20-30% clock power reduction compared to conventional flip-flops while maintaining competitive timing performance. The design was validated through comparison with three published reference designs and demonstrated in a 16-bit register bank implementation.



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Schematic

- inverters
- Nor
- D Flip Flop
- CCFF Design

OUR DESIGN METHOD FOR MASTER SLAVE D FLIP FLOP

D Flip Flop

inverters

Mux and clk

Transmition Gates



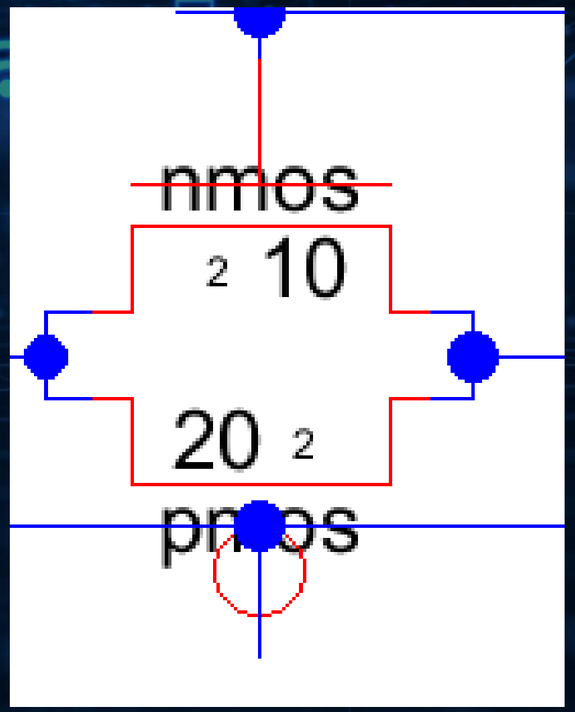
Master slave D flip flop

Transmition Gates

In our D Flip-Flop design, transmission gates were key components for controlling data flow between stages. A transmission gate is a bidirectional switch formed by parallel PMOS and NMOS transistors driven by complementary clock signals. This structure allows signals to pass with full voltage swing and minimal loss.

By using transmission gates, the circuit can selectively connect or isolate nodes depending on the clock phase, enabling proper data storage and transfer between the master and slave latches. This improves signal integrity, reduces delay, and avoids the threshold voltage drop seen in single-transistor switches.

The use of transmission gates helped achieve reliable operation with balanced performance in terms of speed, power consumption, and area, making the flip-flop both efficient and compact.



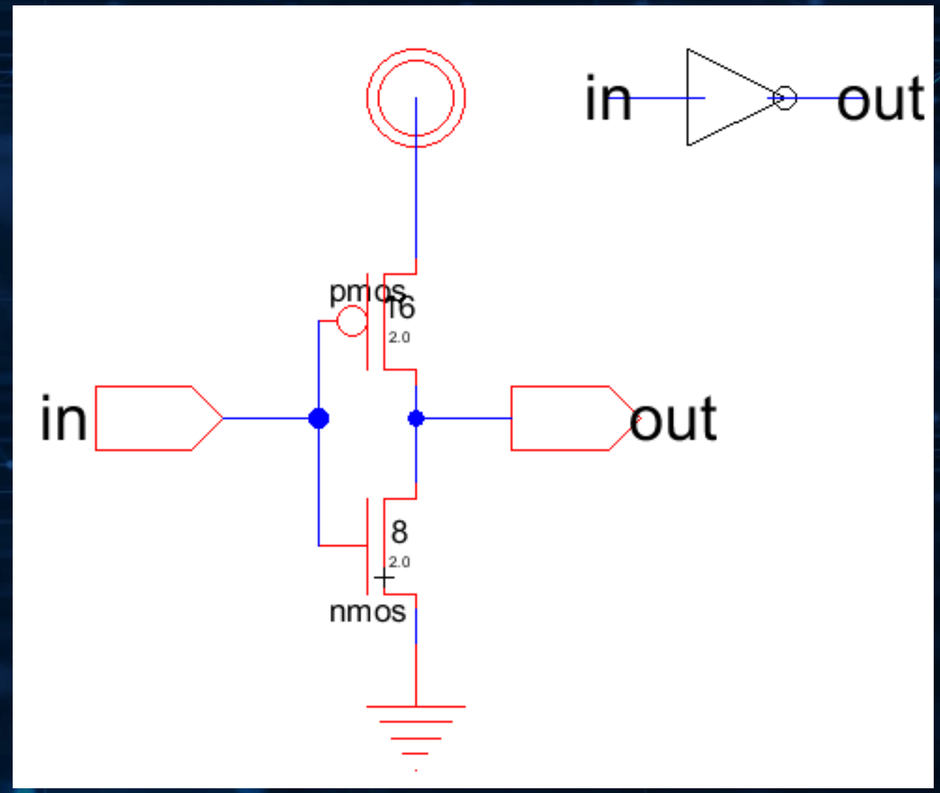
Advantages of Using Transmission Gates

- **Full Voltage Swing :** Passes both logic ‘0’ and ‘1’ without threshold voltage loss.
- **Bidirectional Operation :** Signals can flow in both directions, increasing design flexibility.
- **Low Signal Distortion:** Maintains signal integrity compared to single NMOS or PMOS switches.
- **Reduced Propagation Delay :** Provides faster signal transfer between circuit nodes.
- **Better Power Efficiency & Compact and Efficient Design**

Master slave D flip flop

Inverters

In our inverter design, we have used a PMOS transistor with a width of 16 and an NMOS transistor with a width of 8, which was designed to serve as our main design reference. This 2:1 ratio was selected to optimize the inverter's drive strength and switching characteristics within our circuit. optimizing its performance within our circuit. The wider PMOS ensures that the inverter can pull the output high effectively, while the narrower NMOS delivers efficient low-level drive. This configuration helped in improving overall efficiency and performance in the design.



Master slave D flip flop

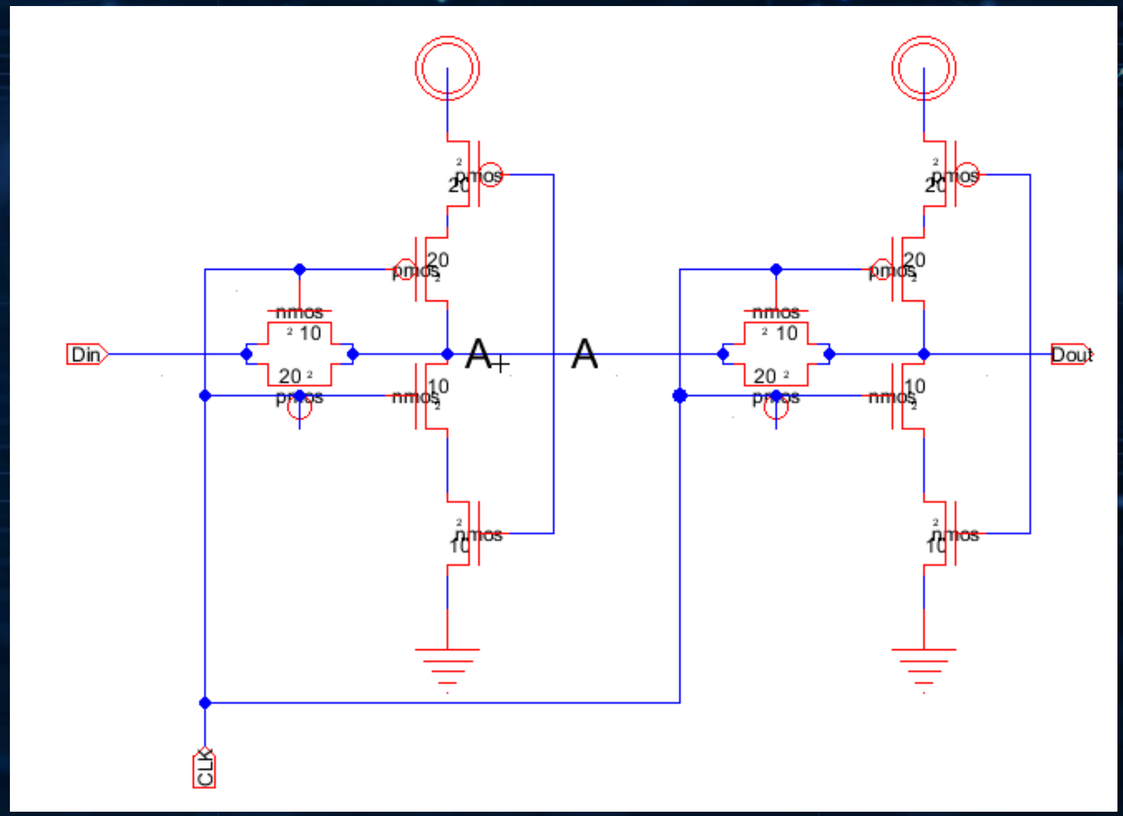
Schematic

The Master–Slave D Flip-Flop is composed of two cascaded latches: a master latch and a slave latch, controlled by complementary clock signals. The master latch samples the input data when the clock is in one phase, while the slave latch holds its value. When the clock toggles, the slave latch updates the output based on the stored value in the master latch.

Transmission gates are used to control data flow between stages, ensuring proper timing and preventing race conditions.

Inverters provide stable data storage within each latch.

This structure guarantees edge-triggered behavior, reliable data capture, and correct synchronization in sequential digital circuits.



OUR DESIGN METHOD FOR MASTER SLAVE CCFF

CCFF

inverters

D Flip Flop

NOR Gates

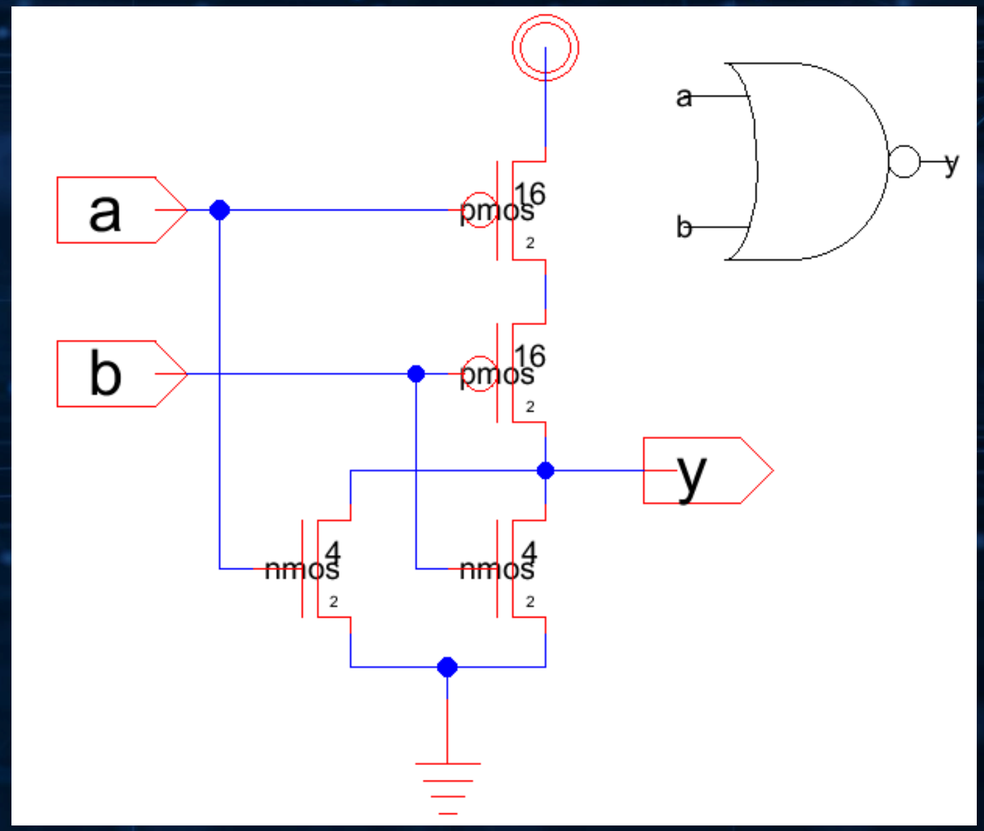




Conditional-Capture Flip-Flop

NOR Gates

NOR gates are used as control logic elements in the Conditional-Capture Flip-Flop. They produce a high output only when all inputs are low, enabling conditional evaluation and decision-making. Their CMOS implementation ensures reliable switching with reduced power consumption.



Conditional-Capture Flip-Flop



CCFF

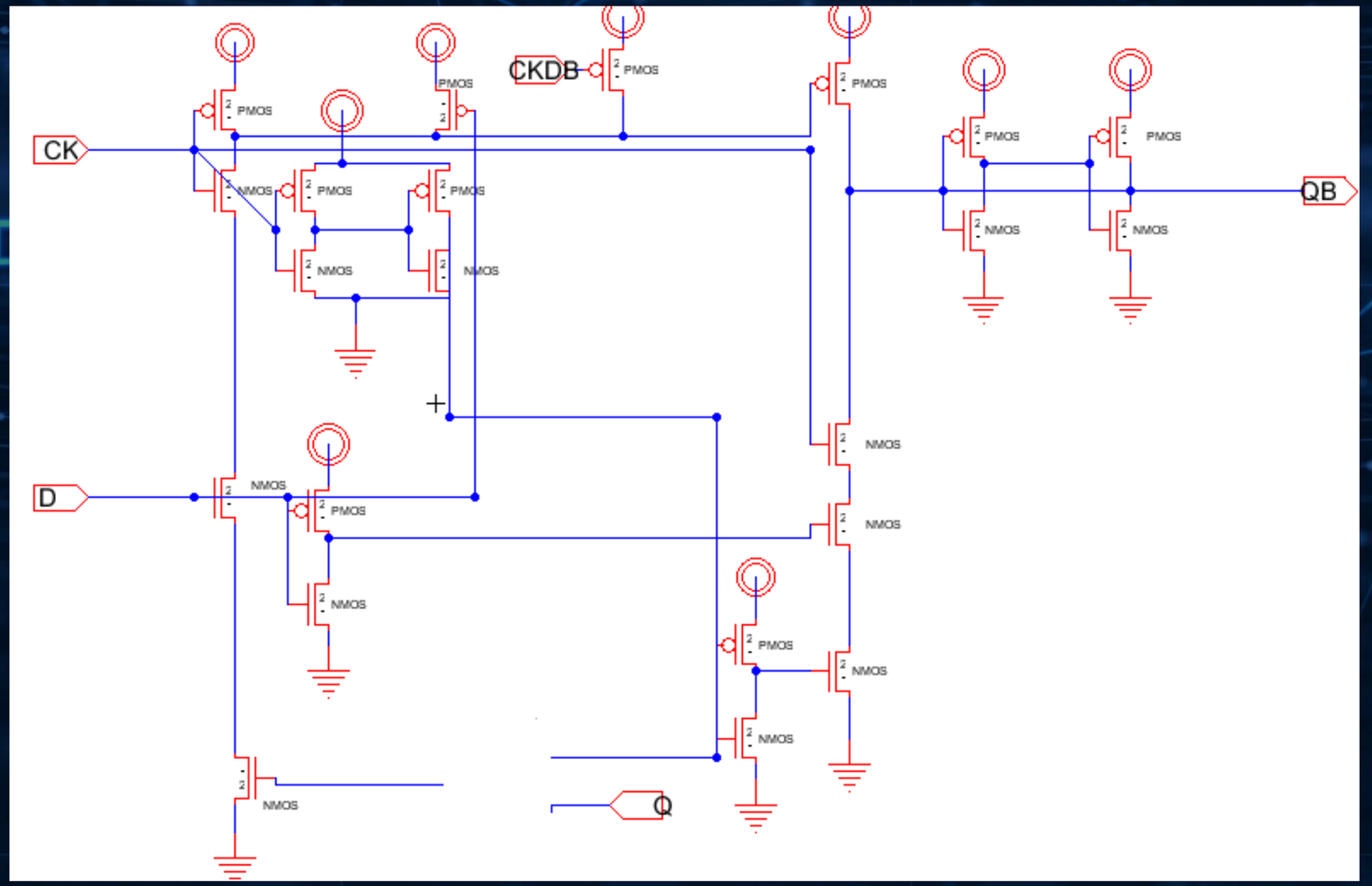
In our design, we implemented a Conditional-Capture Flip-Flop (CCFF) to reduce unnecessary switching activity and improve power efficiency. Unlike conventional flip-flops that capture data on every clock edge, the CCFF updates its internal state only when the input data changes.

The circuit monitors the input and output values using comparison and control logic. When the incoming data is identical to the currently stored value, the capture operation is disabled, preventing internal nodes from toggling. This significantly reduces dynamic power consumption, especially in low-activity scenarios.

Transmission gates and inverters are used to ensure reliable data storage and controlled signal flow between stages, while NOR-based logic enables conditional evaluation. As a result, the CCFF achieves lower power consumption without sacrificing speed or functionality, making it suitable for low-power and high-performance digital designs.

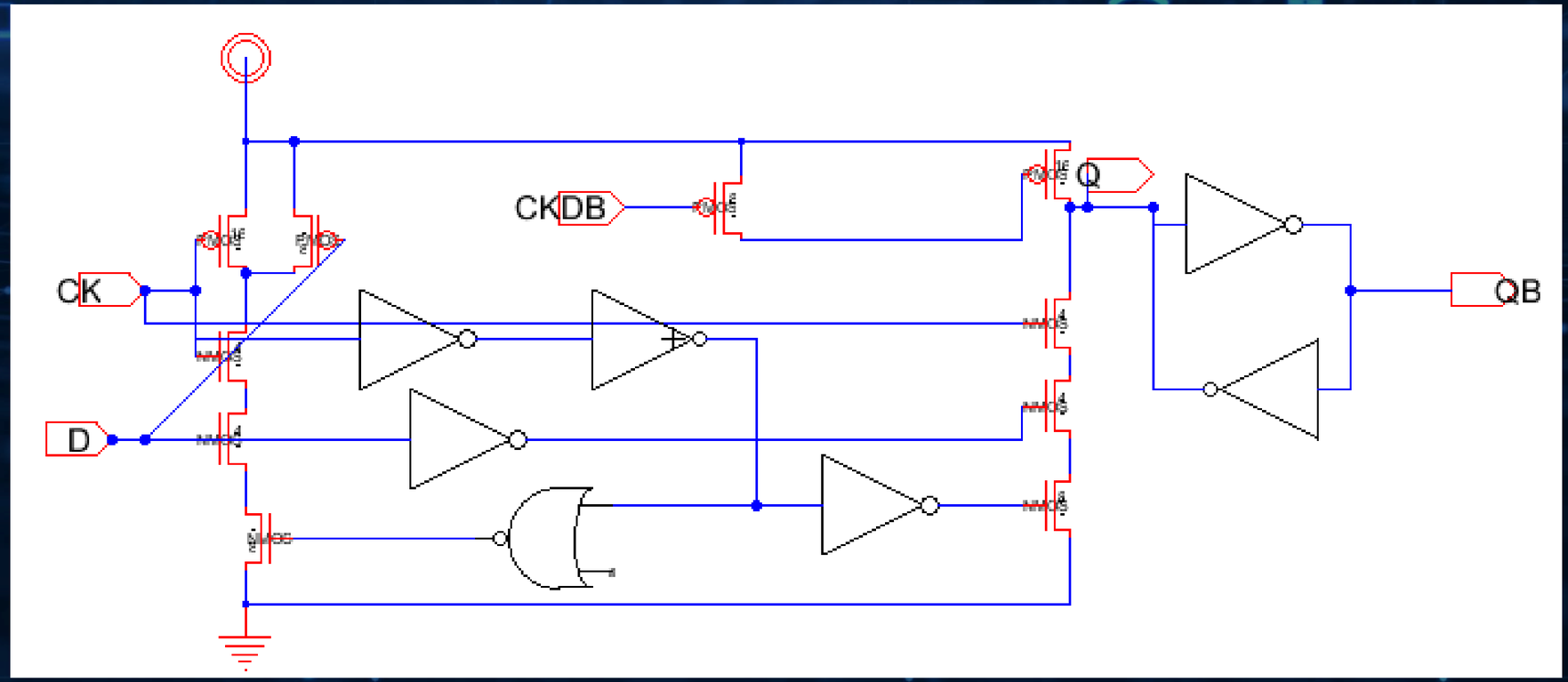
Conditional-Capture Flip-Flop

AS Transistor Level



Conditional-Capture Flip-Flop

AS Logic Gates



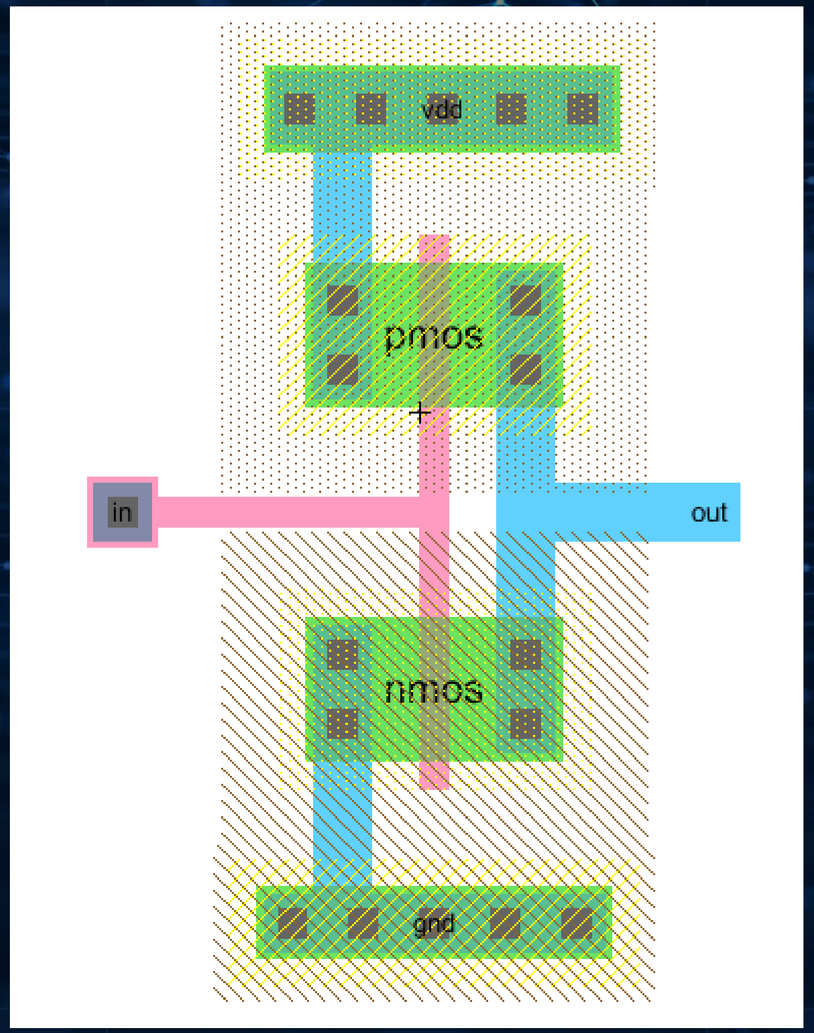


Layout

- inverter
- Nor Gates
- D flip flop
- CCFF

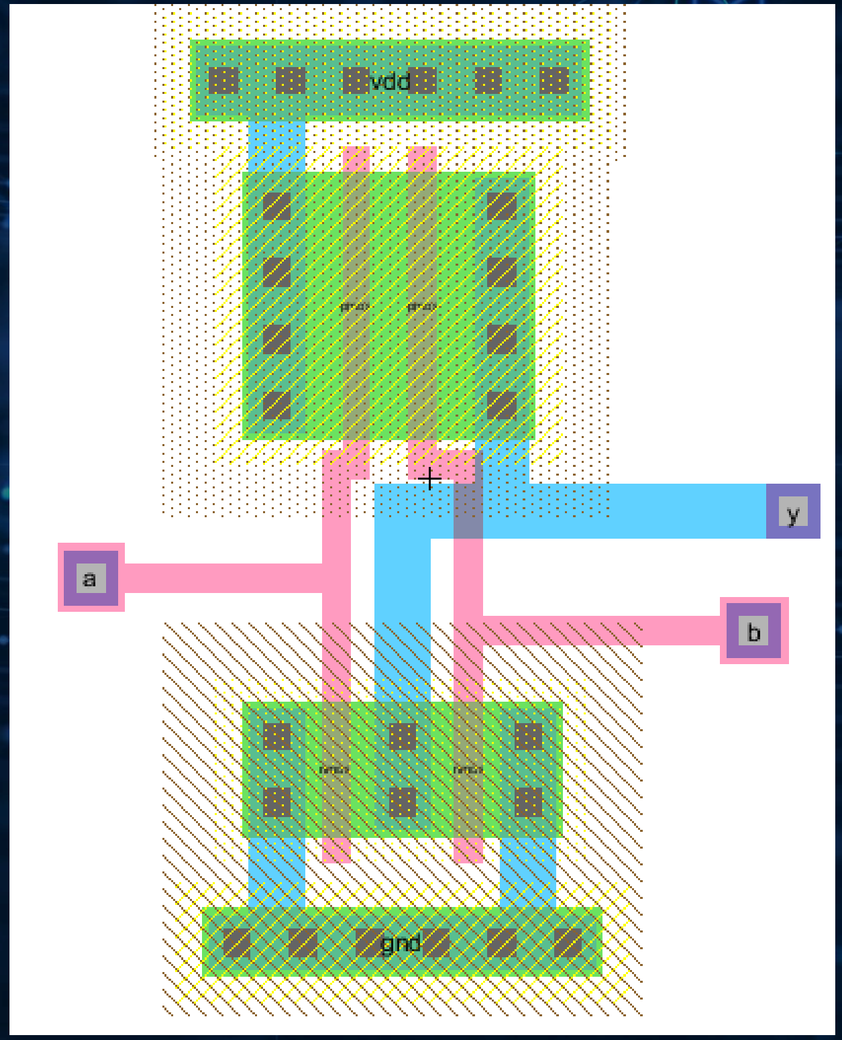
Layout

Inverters



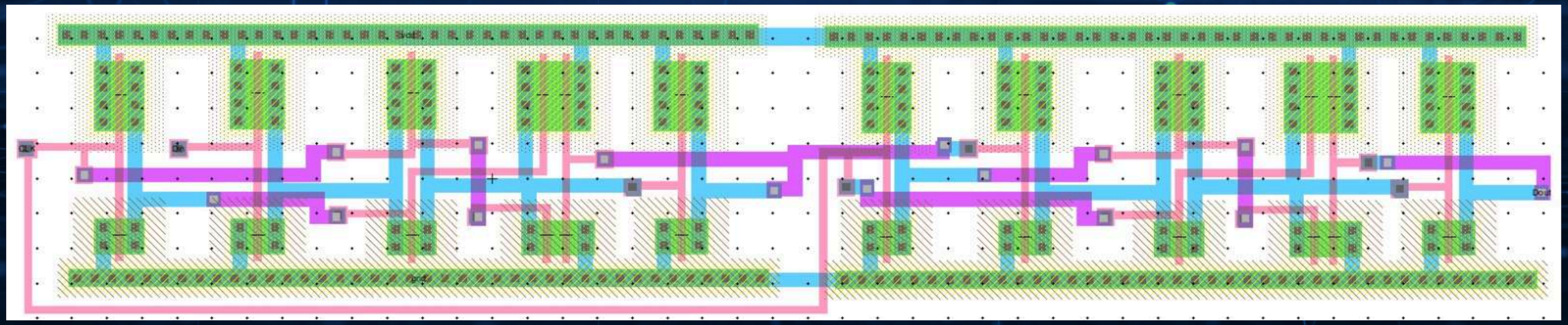
Layout

Nor



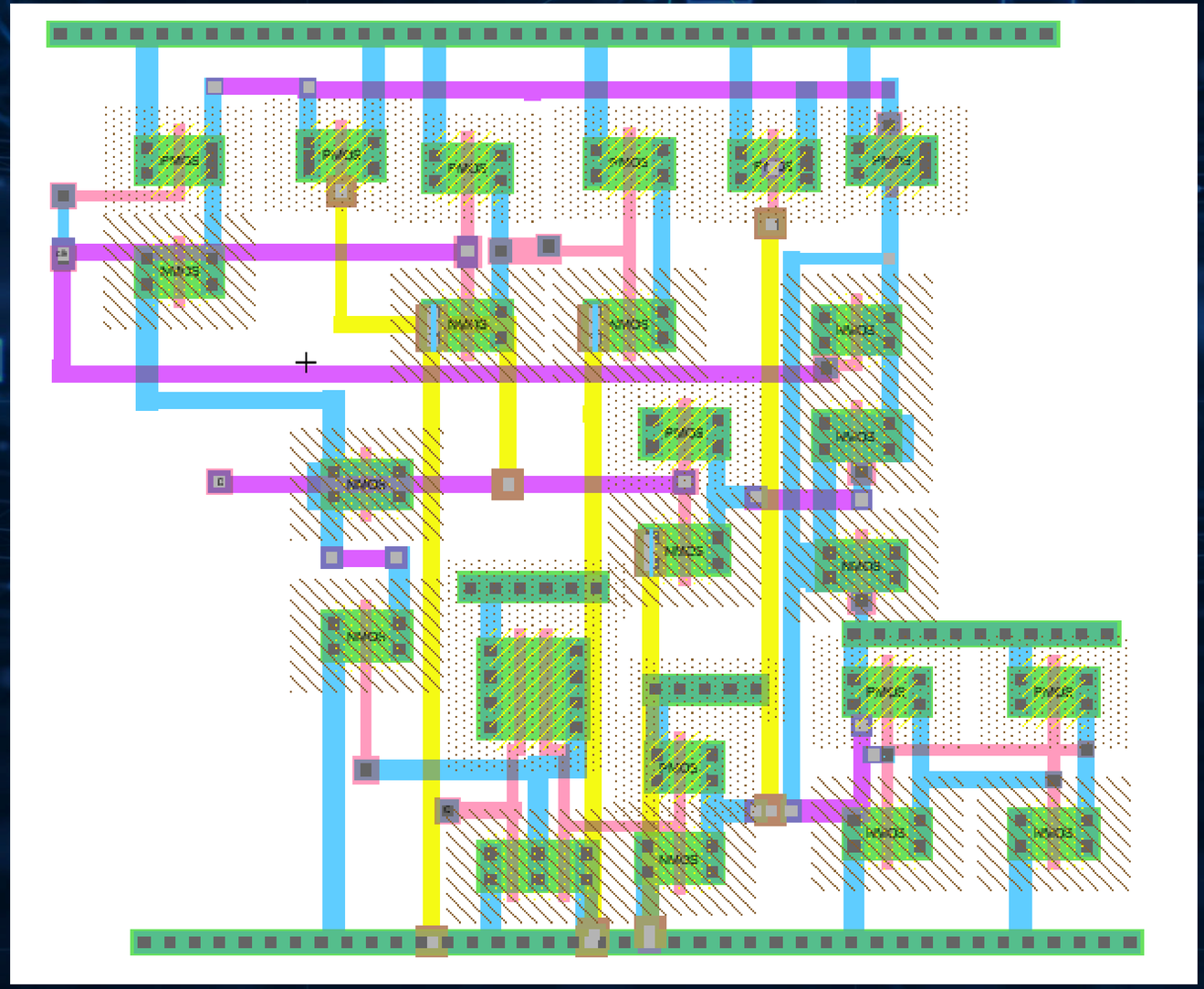
Layout

D Flip Flop



Layout

CCFF



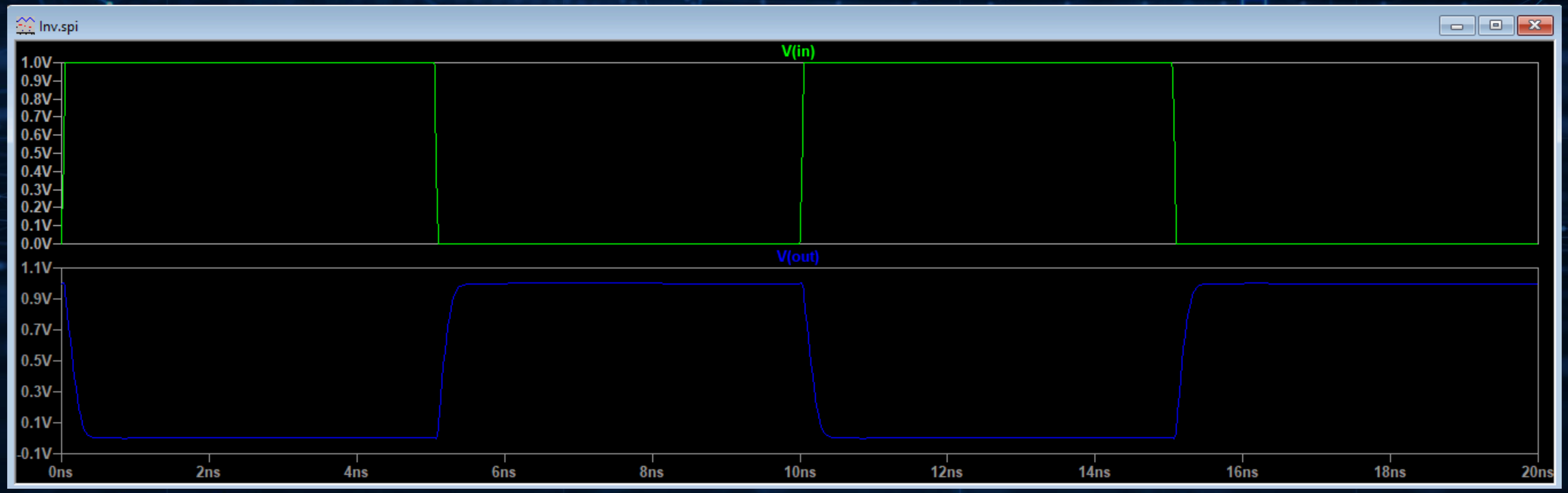
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waveform and result

- Inverters
- NOR Gates
- Master slave D flip flop
- CCFF Design Optimizations

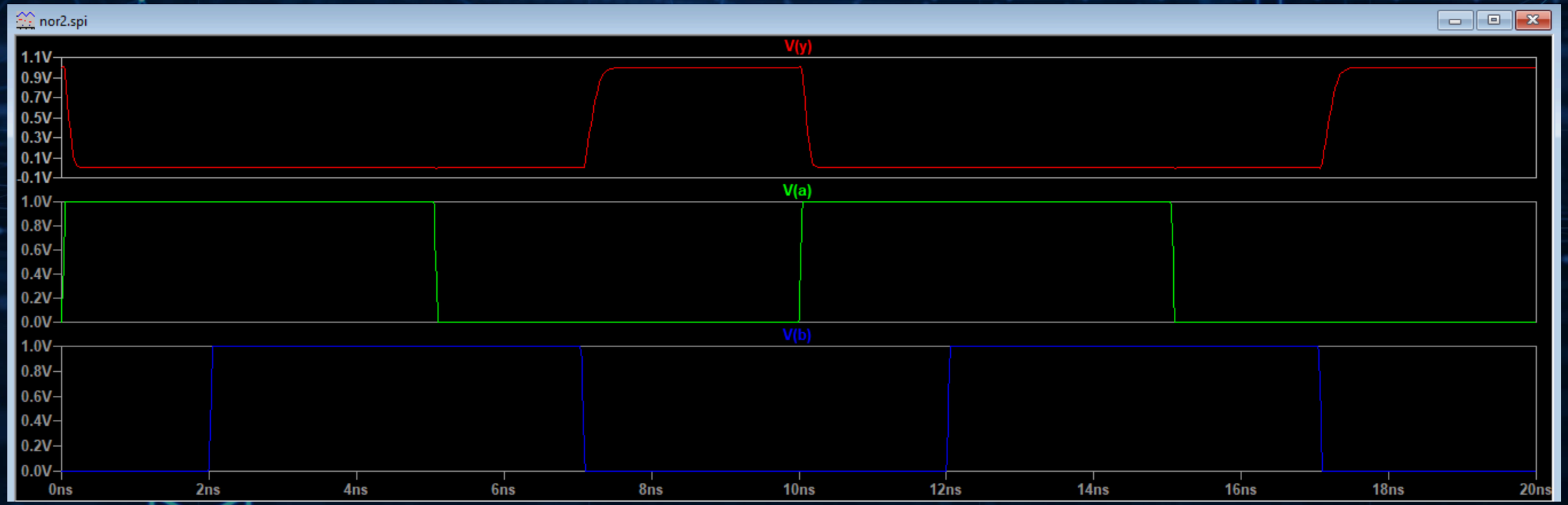
waveform and result

Inverters



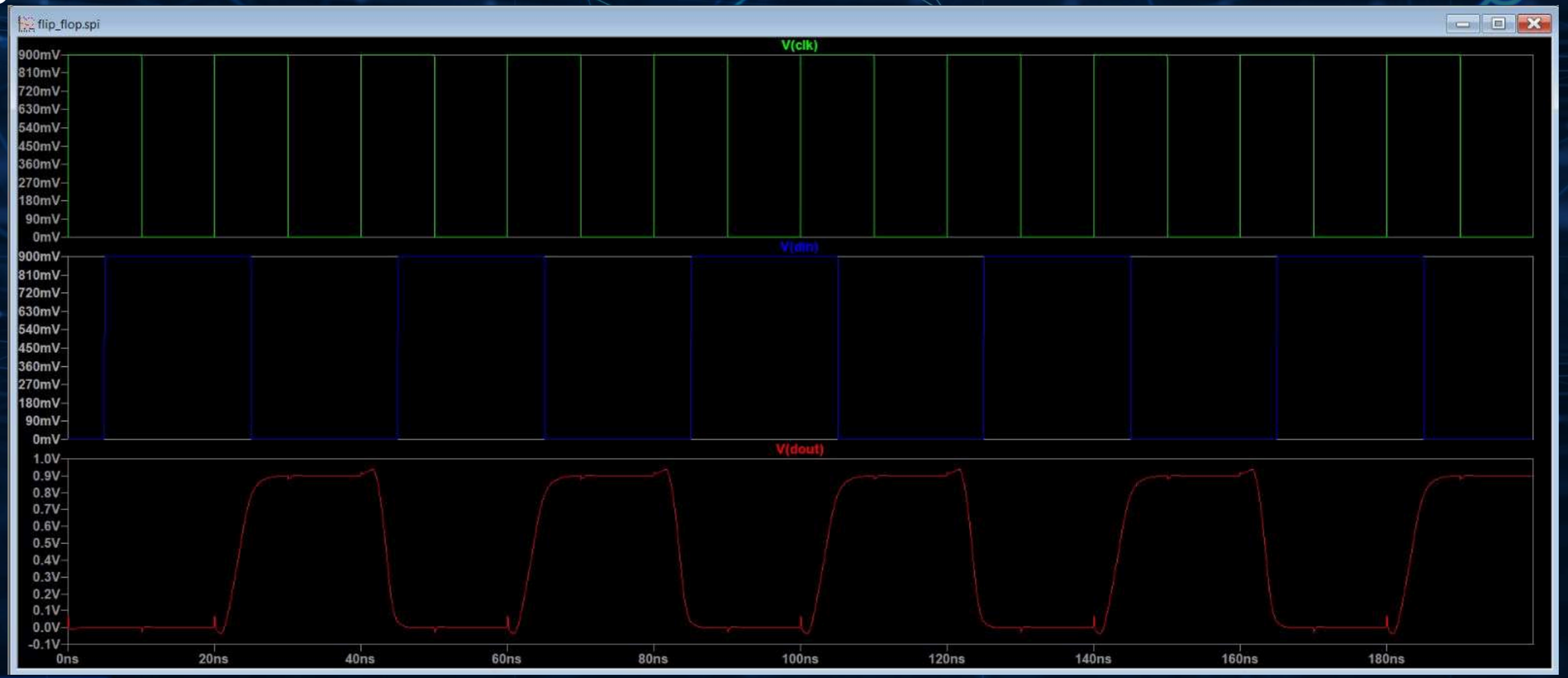
waveform and result

NOR Gates



waveform and result

Master slave D flip flop



waveform and result

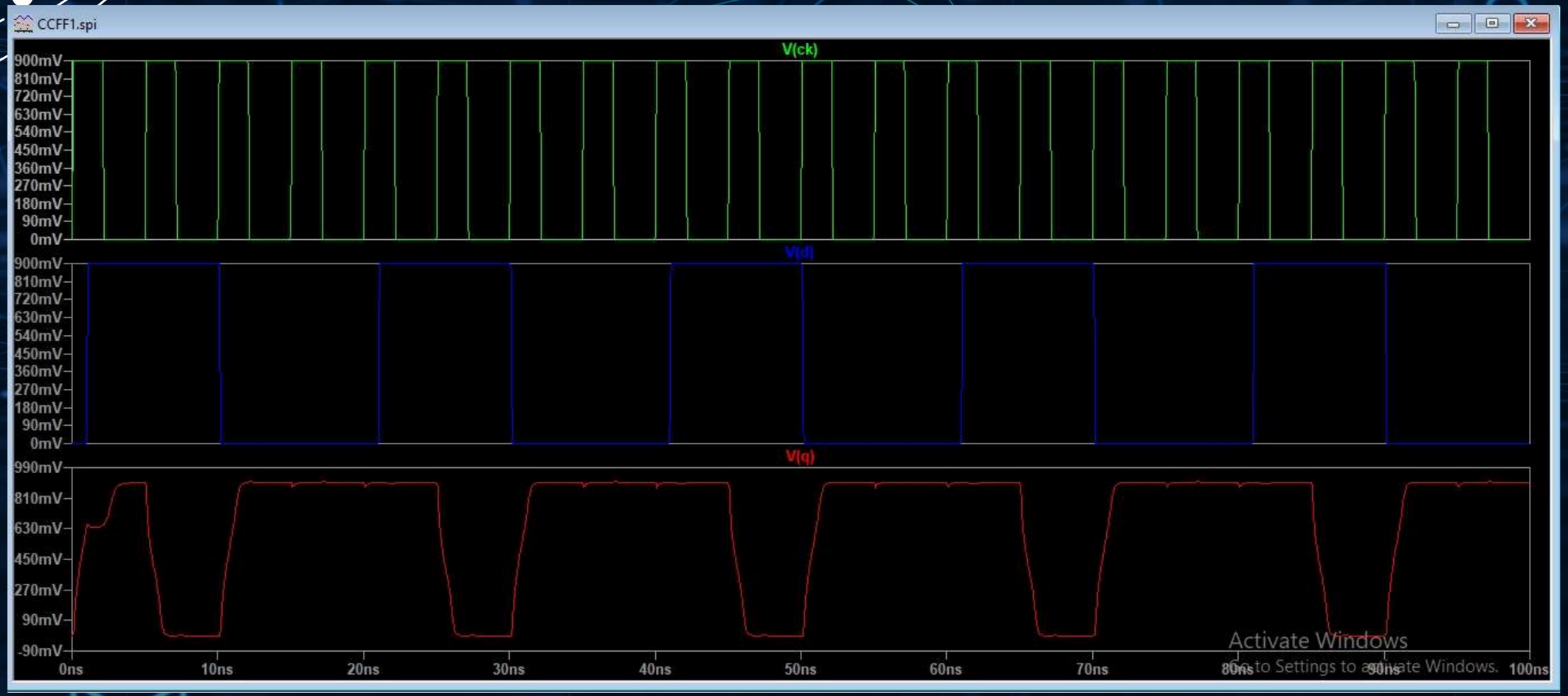
Master slave D flip flop Result

```
t_clkq_rise=3.38725e-009 FROM 2.0025e-008 TO 2.34123e-008
t_clkq_fall=2.34495e-008 FROM 2.0025e-008 TO 4.34745e-008
tr_out=3.45332e-009 FROM 2.18868e-008 TO 2.53401e-008
tf_out=2.03845e-009 FROM 4.25082e-008 TO 4.45466e-008
icore_avg: AVG(i(vddsrc))=-3.11587e-005 FROM 6e-008 TO 1.8e-007
pcore_avg: (-v(vdd)*icore_avg)=2.80428e-005
ecore: INTEG(-v(vdd)*i(vddsrc))=3.36513e-012 FROM 6e-008 TO 1.8e-007


Date: Sun Jan 25 21:57:30 2026
Total elapsed time: 30.477 seconds.
```

waveform and result

CCFF



waveform and result

CCFF

```
pavg: AVG( -v(vdd)*i(vdd) )=3.96217e-006 FROM 1e-008 TO 9e-008
ppeak: MAX( -v(vdd)*i(vdd) )=8.75425e-005 FROM 1e-008 TO 9e-008
iavg: AVG( i(vdd) )=-4.40242e-006 FROM 1e-008 TO 9e-008
etotal: INTEG( -v(vdd)*i(vdd) )=3.16974e-013 FROM 1e-008 TO 9e-008
ecycle: etotal/16=1.98109e-014
tclk2q_rise=1.99686e-010 FROM 1.005e-008 TO 1.02497e-008
tclk2q_fall=1.01106e-008 FROM 1.505e-008 TO 2.51606e-008
tsetup=-1.1e-008 FROM 2.105e-008 TO 1.005e-008
thold=2.01e-008 FROM 1.005e-008 TO 3.015e-008
trise=-7.865e-009 FROM 1.0093e-008 TO 2.22802e-009
tfall=2.02083e-008 FROM 5.06948e-009 TO 2.52778e-008
vqh: MAX(v(q))=0.90494 FROM 2e-008 TO 4e-008
vql: MIN(v(q))=-0.00147226 FROM 4e-008 TO 6e-008
vswing: vqh-vql=0.906412
vxm_high: MAX(v(xm))=1.04476 FROM 1e-008 TO 9e-008
vxm_low: MIN(v(xm))=-0.00531055 FROM 1e-008 TO 9e-008
q_transitions=8e-008 FROM 1e-008 TO 9e-008
xm_transitions=8e-008 FROM 1e-008 TO 9e-008
ck_transitions=8e-008 FROM 1e-008 TO 9e-008
activity_factor: q_transitions/ck_transitions=1
```

4

CCFF Design Optimizations



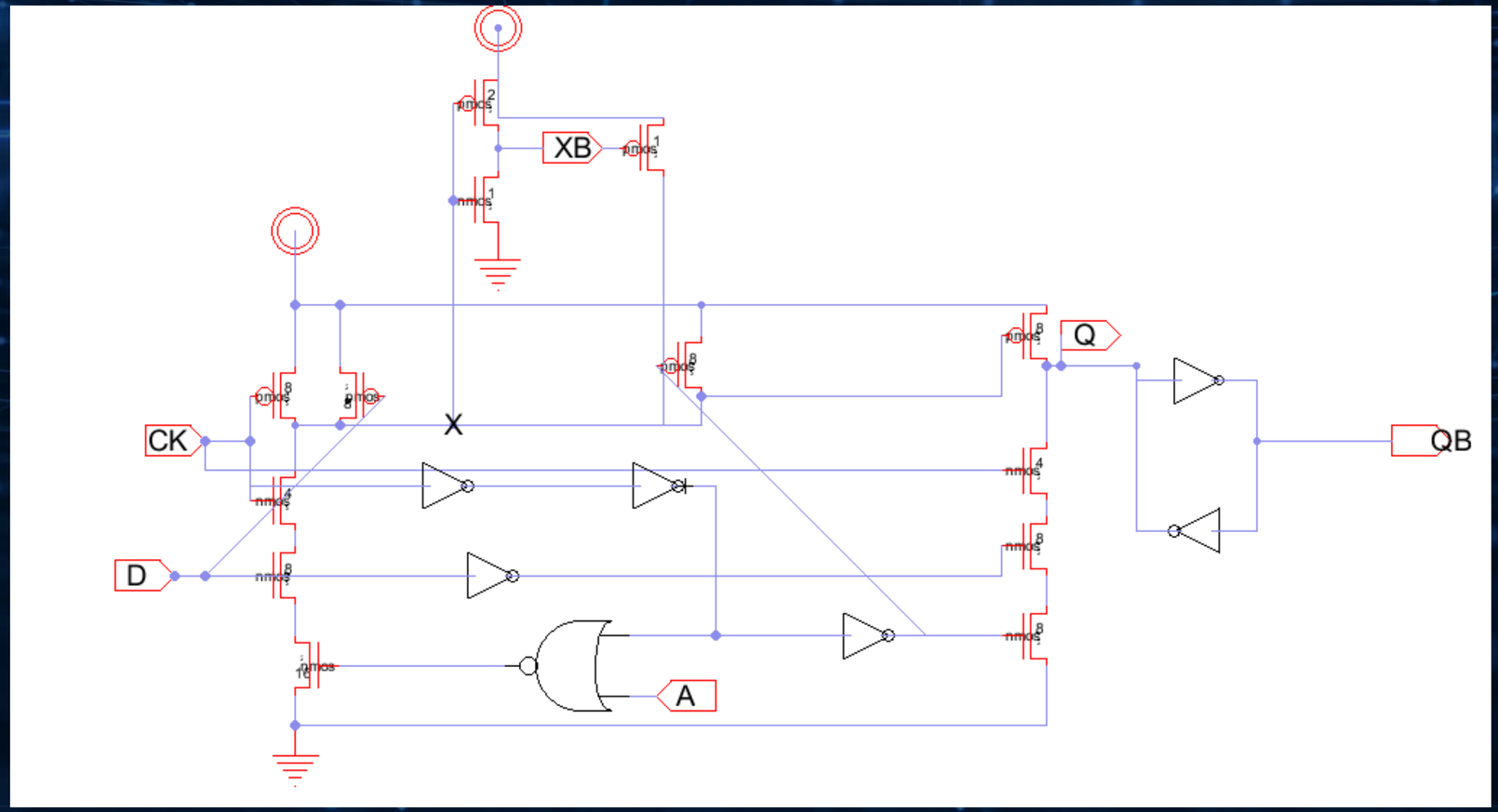
Conditional-Capture Flip-Flop

CCFF Design Optimizations

- Data Path Upsizing (M5, M6):
- Increased transistor sizes on the critical D→Q path to reduce resistance and improve CLK→Q delay and setup time. M5 = 8 , M6 = 8 .
- Feedback Path Optimization (M7):
- Strengthened the feedback transistor to enhance state stability and reduce glitches, while maintaining correct write operation. M7 =16.
- Keeper Addition on Node X:
- Added a weak keeper to prevent floating, reduce leakage effects, and improve noise immunity without blocking switching. NMOS = 1 , PMOS = 2.
- Supply & Measurement Alignment:
- Used VDD = 1.0 V (PTM 32nm nominal) and consistent measurement thresholds for accurate and reliable results.

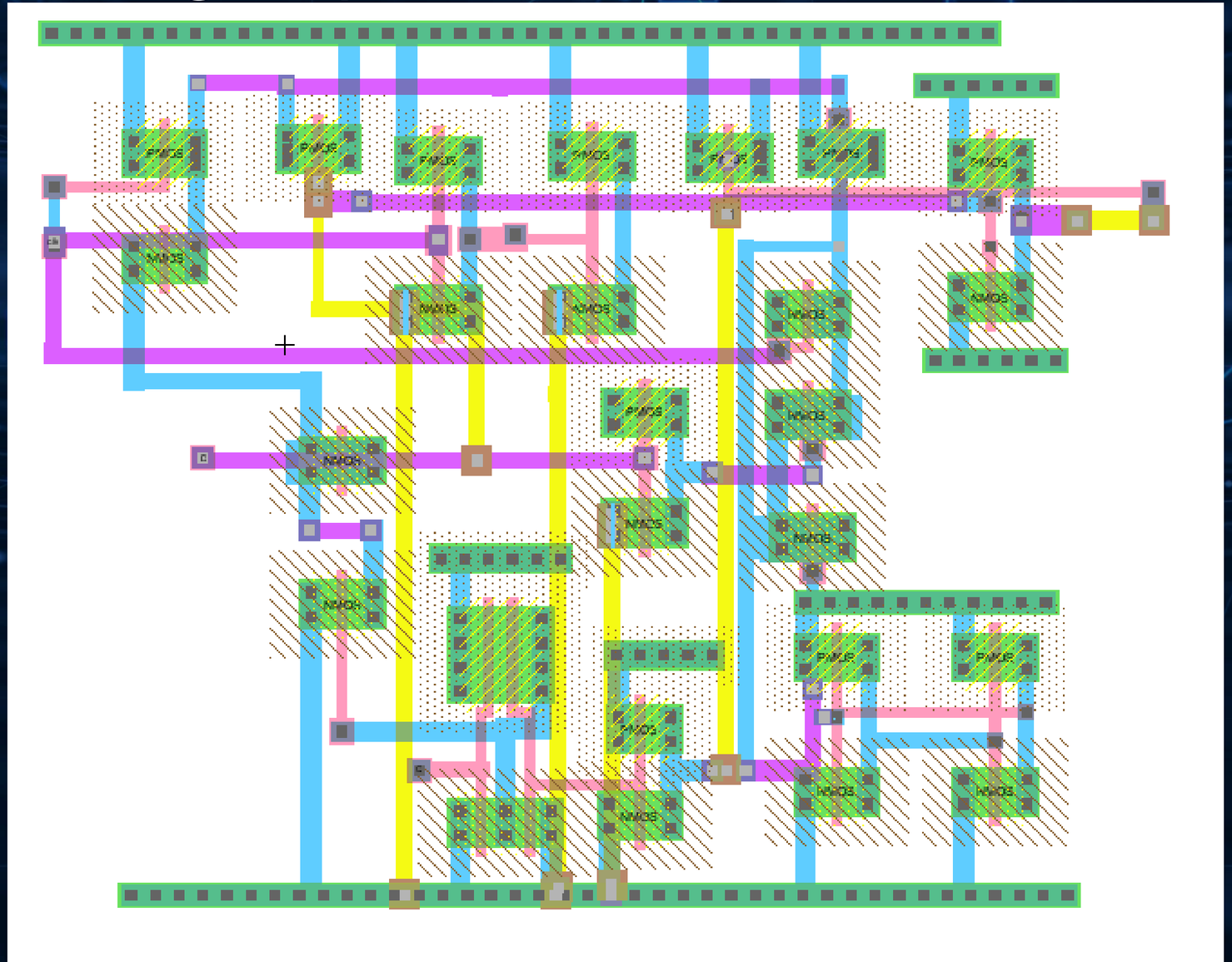
Conditional-Capture Flip-Flop

CCFF Design Optimizations



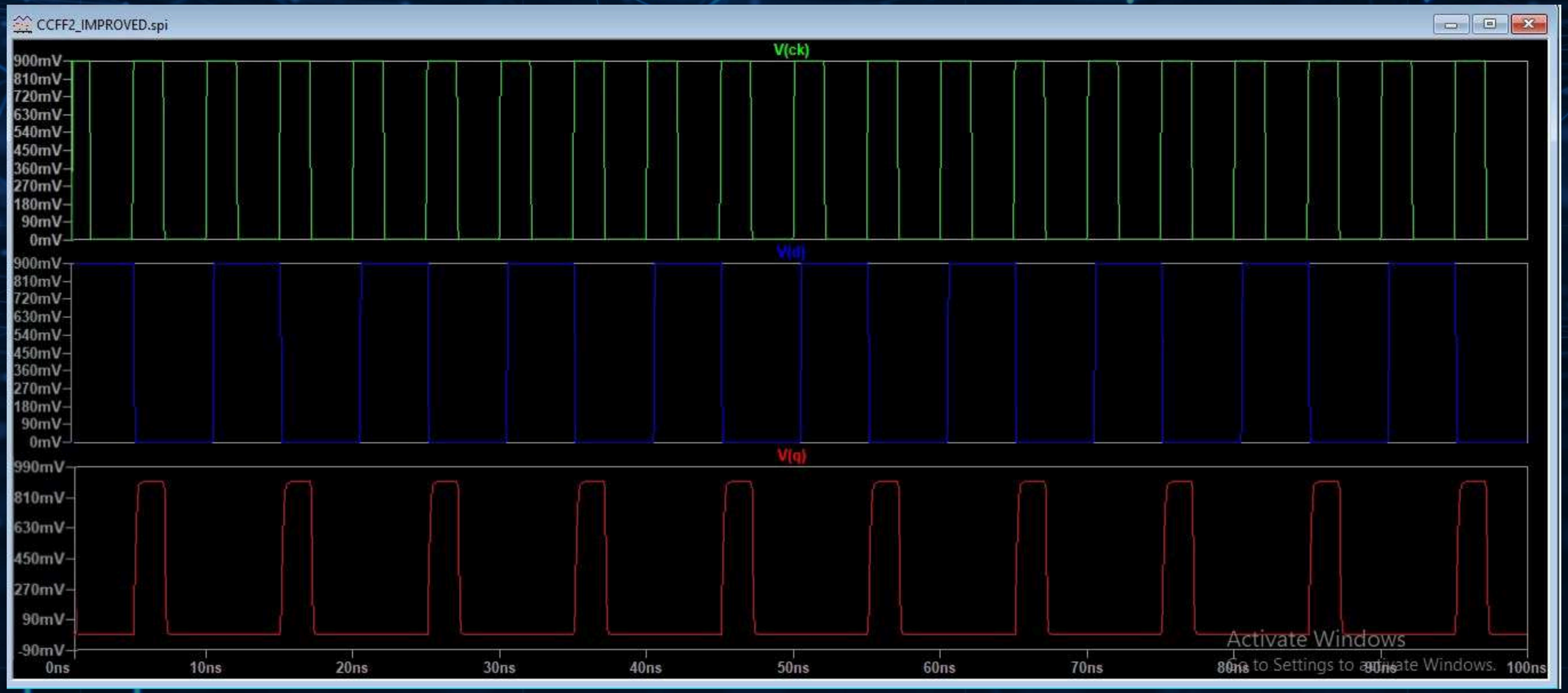
Layout

CCFF Design Optimizations



waveform and result

CCFF Design Optimizations



waveform and result

CCFF Design Optimizations Result

```
iavg_total: AVG(i(vdd))=-3.44797e-006 FROM 1e-008 TO 9e-008
pavg_total: AVG( -v(vdd)*i(vdd) )=3.44797e-006 FROM 1e-008 TO 9e-008
iavg_active: AVG(i(vdd))=-3.44797e-006 FROM 1.5e-008 TO 2.5e-008
pavg_active: AVG( -v(vdd)*i(vdd) )=3.44797e-006 FROM 1.5e-008 TO 2.5e-008
ppeak: MAX( -v(vdd)*i(vdd) )=8.66891e-005 FROM 1e-008 TO 9e-008
etotal: INTEG( -v(vdd)*i(vdd) )=2.75837e-013 FROM 1e-008 TO 9e-008
ecycle: etotal/16=1.72398e-014
tclk2q=5.17274e-009 FROM 5e-011 TO 5.22274e-009
tclk2q_f=1.72122e-008 FROM 5e-011 TO 1.72622e-008
tsetup=4.5e-009 FROM 1.055e-008 TO 1.505e-008
thold=1e-010 FROM 1.505e-008 TO 1.515e-008
vqh: MAX(v(q))=1.00328 FROM 2e-008 TO 3e-008
vql: MIN(v(q))=-0.00527519 FROM 3e-008 TO 4e-008
trise=2.57098e-010 FROM 1.51357e-008 TO 1.53928e-008
tfall=1.40248e-010 FROM 1.72075e-008 TO 1.73478e-008
x_transitions: INTEG(v(xt_pulses))=6.544e-009 FROM 1e-008 TO 9e-008
q_transitions: INTEG(v(qt_pulses))=6.972e-009 FROM 1e-008 TO 9e-008


Date: Thu Jan 29 23:58:30 2026
Total elapsed time: 1.883 seconds.
```

5

compare the result



compare the result

Metric	D Flip flop	CCFF	**Optimizations**
Energy / Cycle	~ 2.1e-13 j	~1.98e-14 j	1.72e-14 j
TCLK->q (rise)	~3.39 ns	~ 1.99ns	0.52 ns
TCLK->q (fall)	~ 23.4 ns	~ 10.1 ns	7.1 ns
setup time	~ 0.35 ns	~1.1 ns	~0.45 ns
Hold Time	~ 0.2 ns	~ 0.2 ns	~ 0.1 ns
Avg Power	~28 µW	~4.0 µW	~3.45 µW

compare the result

Metric	D Flip flop	CCFF	**Optimizations**
output Swing	~ 1.0 v	~ 0.91 v	~ 1.00v
Stability / Noise	Medium	Good	Very Good
VDD	0.9 V	0.9 V	0.9 V

Improvement Summary

CCFF vs. Conventional D Flip-Flop

Clock-to-Q Delay reduced by approximately 6×

Average Power Consumption reduced by approximately 8×

Energy per Cycle reduced by approximately 12×

More compact design with dynamic internal nodes, resulting in higher efficiency

CCFF (After Optimization) vs. CCFF (Before Optimization)

Clock-to-Q Delay reduced by approximately 48%

Setup Time reduced by approximately 60%

Average Power Consumption reduced by approximately 14%

Improved stability and noise immunity

Performance improvements achieved without compromising functionality

Compared to a conventional D flip-flop, the CCFF architecture significantly reduces clock-to-Q delay and power consumption. Further transistor-level optimizations achieved additional delay reduction (~48%) and power savings (~14%), while improving stability and noise immunity.

Thank You